Assignment 2

Hi everyone, to complete Assignment 2 (Project Planning), we need to assign roles. Since we are a team of 6, each of us will need to take on a **Technical Task** and a **Secondary Role** (Management/Documentation/Testing).

1. Team Roles:

Team Member	Preferred Site (Site 1 / Site 2)	Specific Technical Tasks You Want to Lead	Secondary Role (Admin/Doc/Test)
Yiqin Huang	Site 1 (Private Cloud)	Install a Hypervisor	Troubleshooting
El Yazid Sidelkheir	Site 1 (Private Cloud)	Connect to the Public Cloud	Documentation
Justin Ross	Site 1 (Private Cloud)	Configure the Switch	Team Management
Ru Wang	Site 2 (Public Cloud)	Connect to the Private Cloud Network (Site 1)	Team Management
Omer Dengiz	Site 2 (Public Cloud)	Create and Manage VM's	Troubleshooting
Bailey Kulla	Site 2 (Public cloud)	Configure the Switch	Documentation

2. Lab 2 Project Planning Survey

1. Timeline & Availability

- Brainstorming Phase: We need to meet early to discuss value-added options. Which evening this week works best for you?

Team Member	Mon	Tue	Wed	Thu	Fri	Weekend (Sat/Sun)

Yiqin Huang	After 8pm	After 8pm	After 8pm	Flexible	After 8pm	Flexible
El Yazid Sidelkheir	After 8pm	After 8pm	After 6pm	AllDay	After 8pm	Flexible
Justin Ross	Between 12 pm and 2:30 pm		After 1 pm	After 1 pm	After 2 pm	Flexible
Ru Wang	Between 12pm and 4pm After 8 pm	After 8 pm	After 4pm	After 3pm	After 3pm	Flexible
Omer Dengiz	After 8pm	After 8pm	After 8pm	Between 1 pm and 4 pm	After 8pm	Flexible
Bailey Kulla	After 5pm	After 11am	After 5pm	Before 9am/after 4pm	Before 10am/after 3pm	Flexible

Research Time: How many days do you estimate you need for initial research on your technical tasks?

Team Member	Days
Yiqin Huang	3
El Yazid Sidelkheir	3
Ru Wang	3
Justin Ross	3
Omer Dengiz	3
Bailey Kulla	3

- Project Dates:
    - Proposed Start Date: 2026-01-12
    - Proposed End Date (Submission): 2026-03-29

2. Communication Plan

- Where will meetings occur?
    - [ ] MS Teams (Recommended by lab)
- When will meetings occur?
    - [ ] Weekly Sync（every Thursday,Saturday)
    - [ ] Twice a week (Start of week plan + End of week review)
    - [ ] As needed (Milestone-based)
- Meeting Minutes:
    - Who records them? (Fixed): Bailey Kulla
    - Where will they be stored?
        - § [ ] MS Teams Files
        - § [ ] Google Docs / Shared Drive

Note: The detailed Gantt Chart, including task breakdown, assignment, and timeline is provided in the attached file named **"Gantt Chart.xlsx"**. Please refer to the sheet "Gantt-Overall" for the complete schedule.

3. Documentation & Tracking

- How will we track documentation to ensure we are using the correct versions?
    - [ ] Google Docs / Office 365 (Live collaboration)
    - [ ] MS Teams Files

3. Test Plan/Test Log:

Phase	Source (Device)	Destination (Target)	Connectivity	Service / Protocol	Test Command	Expected Result	Result	Comment	Test Date-TBD
1. Site 1 Unit Test	(MSP) Jump Server	192.168.MN.193 (GW)		Network	ping 192.168.MN.193	Reply from Gateway			2026/2/15
1. Site 1 Unit Test	(C1) PC-Client	192.168.MN.1 (GW)		Network	ping 192.168.MN.1	Reply from Gateway			2026/2/15
1. Site 1 Unit Test	(C1) Server-01	192.168.MN.129 (GW)		Network	ping 192.168.MN.129	Reply from Gateway			2026/2/15
1. Site 1 Unit Test	(C1) Web Svr	192.168.MN.161 (GW)		Network	ping 192.168.MN.161	Reply from Gateway			2026/2/15
1. Site 1 Unit Test	(C2) PC-Client	192.168.(MN+1).1 (GW)		Network	ping 192.168.(MN+1).1	Reply from Gateway			2026/2/16
1. Site 1 Unit Test	(C2) Web Svr	192.168.(MN+1).161 (GW)		Network	ping 192.168.(MN+1).161	Reply from Gateway			2026/2/16
2. Site 1 Services	(C1) PC-Client	192.168.MN.163		HTTP Web	curl http://192.168.MN.163	Status 200 OK			2026/2/18
2. Site 1 Services	(C1) PC-Client	192.168.MN.130 (DC)		DNS/AD	nslookup internal.lab	Resolved IP			2026/2/18
2. Site 1 Services	(C1) Server-01	192.168.MN.179 (SAN)		iSCSI	iscsiadm -m discovery	Target IQN Found			2026/2/18

2. Site 1 Services	(C2) Server-01	192.168.(MN+1).179	iSCSI	iscsiadm -m discovery	Target IQN Found	2026/2/19
3. Site 2 Unit Test	(MSP) Mgmt Svr	172.30.MN.17 (GW)	Network	ping 172.30.MN.17	Reply from Gateway	2026/3/5
3. Site 2 Unit Test	(C1) LAN-Server	172.30.MN.49 (GW)	Network	ping 172.30.MN.49	Reply from Gateway	2026/3/5
3. Site 2 Unit Test	(C1) Web Svr	172.30.MN.65 (GW)	Network	ping 172.30.MN.65	Reply from Gateway	2026/3/5
3. Site 2 Unit Test	(C2) LAN-Server	172.30.MN.97 (GW)	Network	ping 172.30.MN.97	Reply from Gateway	2026/3/6
3. Site 2 Unit Test	(C2) Web Svr	172.30.MN.113 (GW)	Network	ping 172.30.MN.113	Reply from Gateway	2026/3/6
4. Integration	(S1) C1_Server	172.30.MN.49 (S2 C1)	VPN Routing	tracert 172.30.MN.49	Traffic via VPN Tunnel	2026/3/20
4. Integration	(S1) C2_Client	172.30.MN.113 (S2 Web)	Cross-Site HTTP	curl 172.30.MN.113	Site 2 Welcome Page	2026/3/20
4. Integration	(S1) Jumpbox	172.30.MN.18 (S2 MSP)	Mgmt Access	ssh admin@172.30.MN.18	Login Prompt	2026/3/22
5. Isolation	(C1) PC-Client	192.168.(MN+1).1 (C2)	VLAN Security	ping 192.168.(MN+1).1	Request Timed Out	2026/3/25

5. Isolation	(C1) PC-Client	172.30.MN.97 (S2 C2)	Routing Security	ping 172.30.MN.97	Destination Unreachable	2026/3/25

4. Hardware & Resource Planning *For those choosing Technical Tasks (VMs/Hypervisors), please estimate the resources needed:*

Site 1:

Group	Service Role	Device Name	OS	vCPU	RAM (GB)	Storage (GB)	Network Interface (NIC & VLAN)
MSP / Mgmt	Jump Server	Jumpbox	Windows Server 2019	2	4	60	1x vNIC (VLAN 99)
MSP / Mgmt	Backup & Monitoring Svr	MSP Mgmt	Windows Server 2019	2	4	60	1x vNIC (VLAN 99)
Company 1	Client Workstation (Win)	C1_PC_Windows	Windows 10 Pro	2	4	60	1x vNIC (VLAN 10)
Company 1	Client Workstation (Linux)	C1_PC_Linux	Ubuntu Desktop 20.04	2	4	40	1x vNIC (VLAN 10)
Company 1	Domain Controller / DNS	C1_Server_01	Windows Server 2019	2	4	60	1x vNIC (VLAN 20)
Company 1	Domain Controller / DNS	C1_Server_02	Windows Server 2019	2	4	60	1x vNIC (VLAN 20)
Company 1	Web Server (DMZ)	C1_Web_Svr	Ubuntu Server 20.04	1	2	40	1x vNIC (VLAN 30)

Company 1	iSCSI Storage Server	C1_Storage	TrueNAS Core	2	8	160	1x vNIC (VLAN 40)
Company 2	Client Workstation (Linux)	C2_PC_Linux	Ubuntu Desktop 20.04	2	4	40	1x vNIC (VLAN 110)
Company 2	Domain Controller / DNS	C2_Server_01	Windows Server 2019	2	4	60	1x vNIC (VLAN 120)
Company 2	Domain Controller / DNS	C2_Server_02	Windows Server 2019	2	4	60	1x vNIC (VLAN 120)
Company 2	Web Server (DMZ)	C2_Web_Svr	Ubuntu Server 20.04	1	2	40	1x vNIC (VLAN 130)
Company 2	iSCSI Storage Server	C2_Storage	TrueNAS Core	2	8	160	1x vNIC (VLAN 140)
TOTAL	Virtual Machines: 12	-	Total:	24	56	900	-

Site 2:

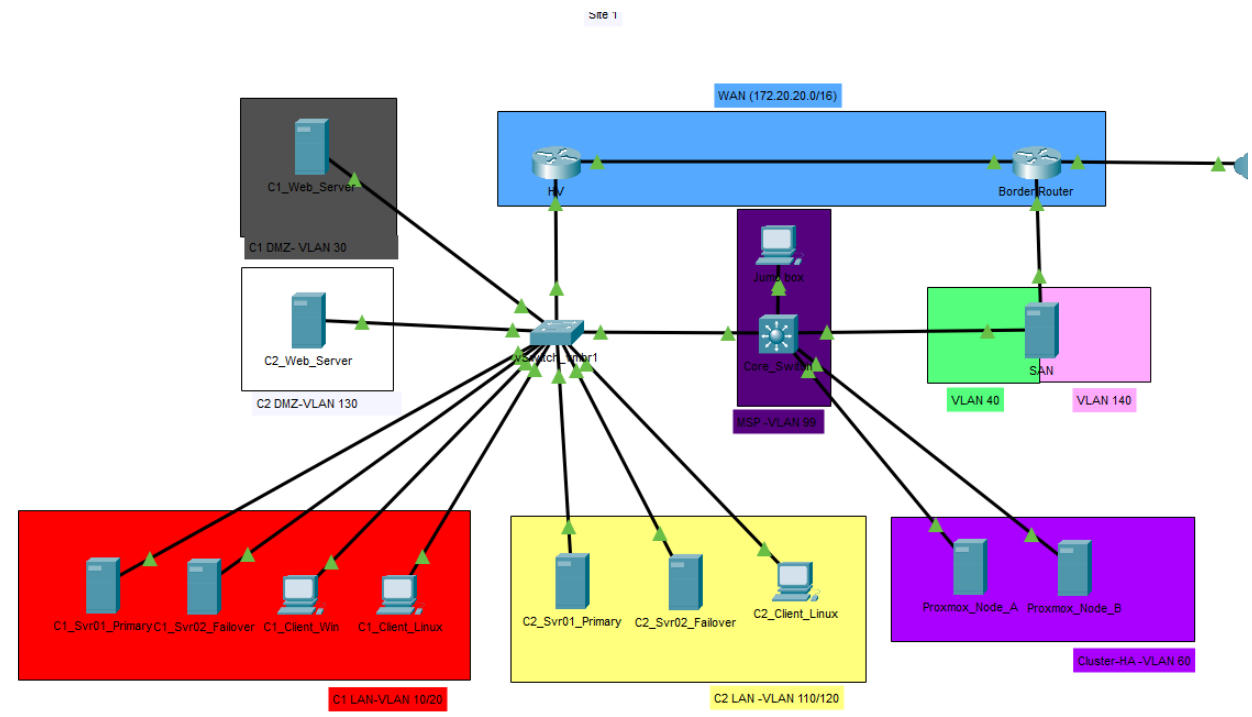
Group	Service Role	Device Name	OS	vCPU	RAM (GB)	Storage (GB)	Network Interface (NIC & VLAN)
MSP / Mgmt	Jump Server (VPN Entry)	Jumpbox	Windows Server 2019	2	4	60	1x vNIC (VLAN 201 - Black)
MSP / Mgmt	Management Server	MSP_Server	Windows Server 2019	2	4	60	1x vNIC (VLAN 202 - Yellow)

MSP / Mgmt	Domain Controller (New)	MSP-DC	Windows Server 2019	2	4	60	1x vNIC (VLAN 202 - Yellow)
Company 1	Web Server (DMZ)	C1_Web_Svr	Ubuntu Server 20.04	1	2	40	1x vNIC (VLAN 130 - Purple)
Company 1	Domain Controller	C1_Server_01	Windows Server 2019	2	4	60	1x vNIC (VLAN 110 - Brown)
Company 1	Client PC	C1_PC_Client	Windows 10 Pro	2	4	60	1x vNIC (VLAN 110 - Brown)
Company 1	iSCSI Storage	C1_Storage	TrueNAS Core	2	8	160	1x vNIC (VLAN 140 - Green)
Company 2	Web Server (DMZ)	C2_Web_Svr	Ubuntu Server 20.04	1	2	40	1x vNIC (VLAN 230 - Pink)
Company 2	Domain Controller	C2_Server_01	Windows Server 2019	2	4	60	1x vNIC (VLAN 210 - Red)
Company 2	Client PC	C2_PC_Client	Ubuntu Desktop 20.04	2	4	40	1x vNIC (VLAN 210 - Red)
Company 2	iSCSI Storage	C2_Storage	TrueNAS Core	2	8	160	1x vNIC (VLAN 240 - Orange)
TOTAL	Virtual Machines: 11	-	Total Resources:	20	48 GB	800 GB	-

5. Based on the dates and research time collected below, the Team Manager must generate a Gantt Chart for the final report.

Gantt Chart as below:

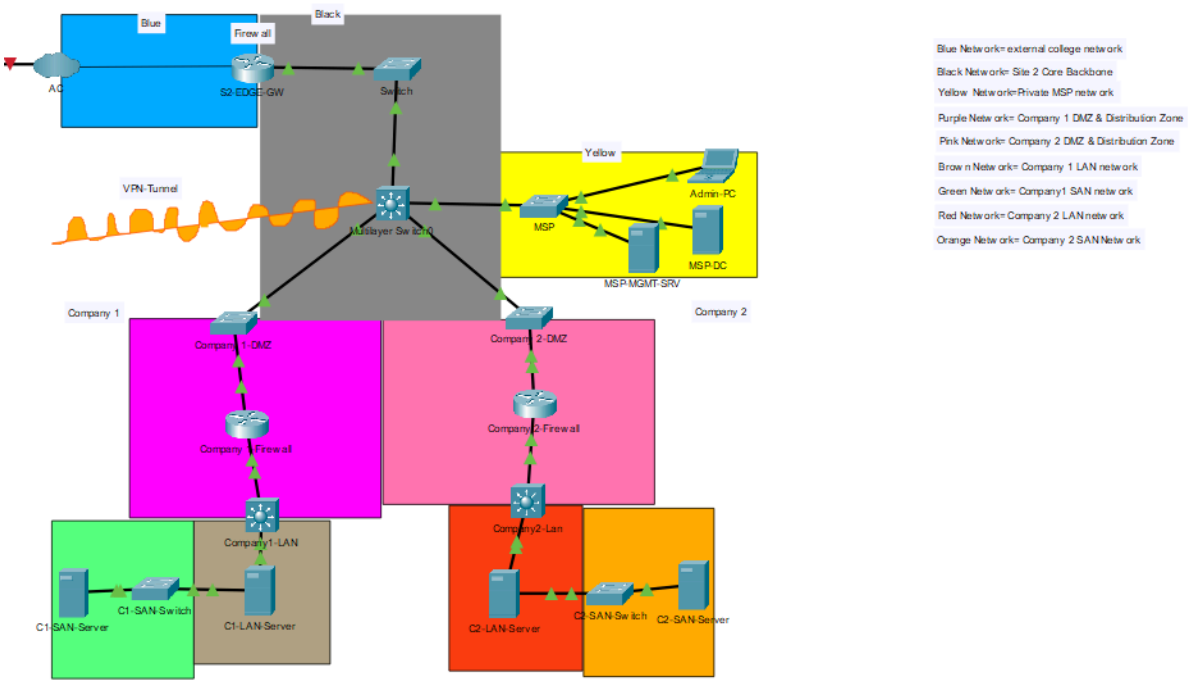
Ni	Phase	Task Name (Specific Technical Task)	Start Da	End Dat	WEEI	Assigned To (Lead	Primary Role (Technical)	Secondary Role (Admin/Doc/Test)	Status
1	INITIATION	Project Kick-off & Setup	1/12	1/18	1	Justin & Ru	N/A	Team Management	100%
1.1	Initiation	Create Team Contract	1/12	1/15	1	Justin Ross	Site 1 Lead	Team Management	100%
1.2	Initiation	Setup MS Teams & Repo	1/12	1/15	1	Ru Wang	Site 2 Lead	Team Management	100%
2	DESIGN	Topology & IP Planning	1/19	1/25	2	Justin & Bailey	Planning Leads	N/A	10%
2.1	Design	Plan S1 IP Subnets & VLANs	1/19	1/21	2	Justin Ross	Plan IP Subnets & Draw Topology	Team Management	10%
2.2	Design	Draw S1 Topology Diagram	1/22	1/25	2	Justin Ross	Plan IP Subnets & Draw Topology	Team Management	10%
2.3	Design	Plan S2 Public Cloud VNet Schema	1/19	1/21	2	Bailey Kulla	Plan IP Addresses & Draw Topology	Documentation	10%
2.4	Design	Draw S2 Cloud Topology Diagram	1/22	1/25	2	Bailey Kulla	Plan IP Addresses & Draw Topology	Documentation	10%
3	BUILD (S1)	Site 1 Infrastructure (Private)	1/26	2/15	3,4,5	Justin & Yiqin	S1 Tech Team	N/A	0%
3.1	Build S1	Config Core Switch (VLANs/Trunk)	1/26	1/29	3	Justin Ross	Plan IP Subnets & Draw Topology	Team Management	0%
3.2	Build S1	Config Core Router (Inter-VLAN)	1/30	2/1	3	Justin Ross	Plan IP Subnets & Draw Topology	Team Management	0%
3.3	Build S1	Install Proxmox Hypervisor	2/2	2/4	5	Yiqin Huang	Install a Hypervisor	Troubleshooting	0%
3.4	Build S1	Config SDN (vSwitch Isolation)	2/5	2/8	5	Yiqin Huang	Install a Hypervisor	Troubleshooting	0%
3.5	Build S1	Deploy S1 VMs (AD/DNS/Storage)	2/9	2/15	5	Yiqin Huang	Install a Hypervisor	Troubleshooting	0%
4	BUILD (S2)	Site 2 Infrastructure (Public)	1/26	2/15	3,4,5	Bailey & Omer	S2 Tech Team	N/A	0%
4.1	Build S2	Create Public Cloud Resource Groups	1/26	1/27	3	Bailey Kulla	Plan IP Addresses & Draw Topology	Documentation	0%
4.2	Build S2	Create VNet (10.53.0.0/16)	1/28	2/1	3	Bailey Kulla	Plan IP Addresses & Draw Topology	Documentation	0%
4.3	Build S2	Deploy Public Cloud VMs (Jump/Web)	2/2	2/8	4	Omer Dengiz	Create and Manage VM's	Troubleshooting	0%
4.4	Build S2	Config NSG Rules (Firewall)	2/9	2/15	5	Omer Dengiz	Create and Manage VM's	Troubleshooting	0%
5	CONNECT	Hybrid Connectivity (VPN)	2/16	2/22	6	El Yazid & Ru	Connect Leads	N/A	0%
5.1	Connect	Config S1 Border VPN (IPSec)	2/16	2/20	6	El Yazid	Connect to the Public Cloud	Documentation	0%
5.2	Connect	Deploy Public Cloud VPN Gateway	2/16	2/20	6	Ru Wang	Connect to the Private Cloud	Team Management	0%
5.3	Connect	MILESTONE: Site-to-Site Ping	2/21	2/22	6	Yiqin / Omer	N/A	Troubleshooting	0%
5.4	Connect	Automated Backups (Veeam)	2/21	2/22	6	Yiqin Huang	Install Backup Agents	Documentation	0%
6	VALUE ADD	Enhancements & Security	2/23	3/1	7,8	All	Value Added	N/A	0%
6.1	Value Add	Site 1: Research & Deploy (TBD)	2/23	3/1	7,8	Justin, Yiqin, El	Value Add Implementation	Doc & Test	0%
6.2	Value Add	Site 2: Research & Deploy (TBD)	2/23	3/1	7,8	Ru,Bailey, Omer	Value Add Implementation	Doc & Test	0%
7	INTEGRATE	System Integration & Fixes	3/2	3/15	8	All	Integration	Troubleshooting	0%
7.1	Integrate	Domain Join (Hybrid Identity)	3/2	3/6	8	Omer / Yiqin	Integration Task	Troubleshooting	0%
7.2	Integrate	Hybrid DNS Forwarding Setup	3/5	3/6	8	Justin / Yiqin	Integration Task	Troubleshooting	0%
7.3	Integrate	Full System Troubleshooting	3/9	3/15	9	Yiqin / Omer	N/A	Troubleshooting	0%
8	CLOSE	Documentation & Submission	3/16	3/29	10	Doc Team	N/A	Documentation	0%
8.1	Closing	Compile Final Technical Report	3/16	3/24	10	El / Bailey	N/A	Documentation	0%
8.2	Closing	Final Review & Submission	3/25	3/29	11	All	N/A	Team Management	0%



6. Network Topology

Site 1:

SIte 2:



7. Site 1 Network info and Site 1 IP Address

VLAN ID	Role	Subnet	IP Range	capacity	Purpose	Color	Company
10	C1-Clients	192.168.MN.0 /25	.1 - .126	126	Staff PC (Windows), printer, WiFi	Red (LAN)	1
20	C1-Servers	192.168.MN.128 /27	.129 - .158	30	AD, DNS, DHCP, File Server (HA)	Red (LAN)	1
30	C1-DMZ	192.168.MN.160 /28	.161 - .174	14	Web Server, Reverse Proxy	Black (DMZ)	1
40	C1-iSCSI	192.168.MN.176 /28	.177 - .190	14	SAN Storage Traffic (Isolated)	Green (SAN)	1
99	MSP-Mgmt	192.168.MN.192 /28	.193 - .206	14	Jumpbox, Switch Mgmt, IPMI	Purple (MSP)	1
60	Cluster-HA	192.168.MN.208 /29	.209 - .214	6	Proxmox Migration / Heartbeat	Purple (Infra)*	1
-	Reserved	192.168.MN.216 /...	.217 - .254	38	Future Expansion	Grey (Reserved)	1
VLAN ID	**Role**	**Subnet**	**IP Range**	**capacity**	**Purpose**	**Color**	**Company**
110	C2-Clients	192.168.(MN+1).0 /25	.1 - .126	126	Staff PC (Linux), printer, WiFi	Yellow (LAN)	2
120	C2-Servers	192.168.(MN+1).128 /27	.129 - .158	30	AD/LDAP, DNS, File Server	Yellow (LAN)	2
130	C2-DMZ	192.168.(MN+1).160 /28	.161 - .174	14	Web Server	White (DMZ)	2
140	C2-iSCSI	192.168.(MN+1).176 /28	.177 - .190	14	SAN Storage Traffic (Isolated)	Magenta (SAN)*	2
-	Reserved	192.168.(MN+1).192 /26	.193 - .254	62	Future Expansion	Grey (Reserved)	2
Interface	**Link Type**	**Network / IP**	**Description**	**Color**			
Gi1/0/24	L3 Uplink	172.20.20.0 /16	WAN / Uplink to Border Router	Blue (WAN)			

Purpose	Device	Role	VLAN	IP Address	subnet	Gateway	DNS	Color
MSP Management Zone	Core Switch	Gateway	99	192.168.MN.193	/25 (.128)	-	-	Purple (MSP)
MSP Management Zone	Jump Server	Jumpbox	99	192.168.MN.194	/25 (.128)	192.168.MN.193	192.168.MN.130	Purple (MSP)
MSP Management Zone	Proxmox Host	Mgmt IP	99	192.168.MN.195	/25 (.128)	192.168.MN.193	192.168.MN.130	Purple (MSP)
MSP Management Zone	SAN	Web Mgmt	99	192.168.MN.196	/25 (.128)	192.168.MN.193	-	Purple (MSP)
MSP Management Zone	Core Switch	Mgmt IP	99	192.168.MN.197	/25 (.128)	192.168.MN.193	-	Purple (MSP)
Company 1	Core Switch	C1-Client GW	10	192.168.MN.1	/25 (.128)	-	-	Red (LAN)
Company 1	PC-Client Pool(Win & Linux)	Staff(DHCP)	10	192.168.MN.2 - .126	/25 (.128)	192.168.MN.1	192.168.MN.130	Red (LAN)
Company 1	Core Switch	C1-Server GW	20	192.168.MN.129	/27 (.224)	-	-	Red (LAN)
Company 1	Server-01(Primary)	AD/DNS/DHCP/File	20	192.168.MN.130	/27 (.224)	192.168.MN.129	127.0.0.1	Red (LAN)
Company 1	Server-02(Secondary/Failover)	AD/DNS/DHCP/File	20	192.168.MN.131	/27 (.224)	192.168.MN.129	192.168.MN.130	Red (LAN)
Company 1	Server-Reserved	Future Servers	20	192.168.MN.132 - .158	-		-	Red (LAN)
Company 1	Core Switch	C1-DMZ GW	30	192.168.MN.161	/28 (.240)	-	-	Black (DMZ)
Company 1	Web Server	WWW-01	30	192.168.MN.162	/28 (.240)	192.168.MN.161	192.168.MN.130	Black (DMZ)
Company 1	Web Server-Reserved	Future Web Servers	30	192.168.MN.163 - .174	-		-	Black (DMZ)
Company 1	Core Switch	C1-iSCSI GW	40	192.168.MN.177	/28 (.240)	-	-	Green (SAN)
Company 1	Storage	iSCSI Target	40	192.168.MN.178	/28 (.240)	192.168.MN.177	empty or 192.168.MN.177	Green (SAN)
Company 1	Storage-Reserved	Future Storage	40	192.168.MN.179-190	/28 (.240)			Green (SAN)
Company 2	Core Switch	C2-Client GW	110	192.168.(MN+1).1	/25 (.128)	-	-	Yellow (LAN)
Company 2	Linux PC-Client Pool	Staff-Linux(DHCP)	110	192.168.(MN+1).2-126	/25 (.128)	192.168.(MN+1).1	192.168.(MN+1).130	Yellow (LAN)
Company 2	Core Switch	C2-Server GW	120	192.168.(MN+1).129	/27 (.224)	-	-	Yellow (LAN)
Company 2	Server-01(Primary)	LDAP/DHCP/File	120	192.168.(MN+1).130	/27 (.224)	192.168.(MN+1).129	127.0.0.1	Yellow (LAN)
Company 2	Server-02(Secondary/Failover)	LDAP/DHCP/File	120	192.168.(MN+1).131	/27 (.224)	192.168.(MN+1).129	192.168.(MN+1).130	Yellow (LAN)
Company 2	Server-Reserved	Future Servers	120	192.168.(MN+1).132 - .158	-		-	Yellow (LAN)
Company 2	Core Switch	C2-DMZ GW	130	192.168.(MN+1).161	/28 (.240)	-	-	White (DMZ)
Company 2	Web Server	WWW-02	130	192.168.(MN+1).162	/28 (.240)	192.168.(MN+1).161	192.168.(MN+1).130	White (DMZ)
Company 2	Web Server-Reserved	Future Web Servers	130	192.168.(MN+1).163 - .174	-		-	White (DMZ)
Company 2	Core Switch	C2-iSCSI GW	140	192.168.(MN+1).177	/28 (.240)	-	-	Magenta (SAN)*
Company 2	Storage	iSCSI Target	140	192.168.(MN+1).178	/28 (.240)	192.168.(MN+1).177	empty or192.168.(MN+1).177	Magenta (SAN)*
Company 2	Storage-Reserved	Future Storage	140	192.168.(MN+1).179-.190	/28 (.240)		-	Magenta (SAN)*

Purpose	Device	Interface	VLAN	IP Address	subnet	Gateway	DNS	Color
Cluster-HA	Proxmox Node A	eth1 (Cluster)	60	192.168.MN.209	/29 (.248)	192.168.MN.193	192.168.MN.130	Purple (Infra)*
Cluster-HA	Proxmox Node B	eth1 (Cluster)	60	192.168.MN.210	/29 (.248)	192.168.MN.193	192.168.MN.130	Purple (Infra)*

8. Site 2 Network info and Site 2 IP Address

VLAN ID	Role	Subnet	IP Range	capacity	Purpose	Color	Company
201	Backbone	172.30.MN.0 /29	.1 - .6	6	Router & FW Interconnects	Black (Core)	MSP
202	MSP-Mgmt	172.30.MN.16 /28	.17 - .30	14	MSP Admin PC & Servers	Yellow (MSP)	MSP
-	Reserved	172.30.MN.32 /28	.33 - .46	14	Future Expansion	Grey (Reserved)	MSP
110	C1-LAN	172.30.MN.48 /28	.49 - .62	14	Staff PC (Windows), Printers	Brown (LAN)	1
130	C1-DMZ	172.30.MN.64 /28	.65 - .78	14	Web Server (Public Access)	Purple (DMZ)	1
140	C1-SAN	172.30.MN.80 /28	.81 - .94	14	SAN Storage Traffic	Green (SAN)	1
VLAN ID	Role	Subnet	IP Range	capacity	Purpose	Color	Company
210	C2-LAN	172.30.MN.96 /28	.97 - .110	14	Staff PC (Linux), Printers	Red (LAN)	2
230	C2-DMZ	172.30.MN.112 /28	.113 - .126	14	Web Server (Public Access)	Pink (DMZ)	2
240	C2-SAN	172.30.MN.128 /28	.129 - .142	14	SAN Storage Traffic	Orange (SAN)	2
240	C2-SAN	172.30.MN.240 /28	.241 - .254	14	SAN Storage Traffic (Isolated)	Orange (SAN)	2
-	Reserved	172.30.MN.255 /32	-	-	Broadcast / Unused	Grey	-

Interface	Link Type	Network / IP	Description	Color
Gi0/0	L3 Uplink	DHCP (Public)	WAN / Uplink to College Network	Blue (WAN)

Purpose	Device	Role	VLAN	IP Address	subnet	Gateway	DNS	Color
MSP Backbone	**S2-EDGE-GW**	Edge Router	201	172.30.MN.1	/29 (.248)	(ISP IP)	8.8.8.8	Black
MSP Backbone	**Multilayer Switch0**	Core Switch	201	172.30.MN.2	/29 (.248)	172.30.MN.1	172.30.MN.1	Black
MSP Mgmt Zone	**Multilayer Switch0**	MSP-GW	202	172.30.MN.17	/28 (.240)	-	-	Yellow
MSP Mgmt Zone	**MSP-MGMT-SRV**	Admin Server	202	172.30.MN.18	/28 (.240)	172.30.MN.17	172.30.MN.17	Yellow
MSP Mgmt Zone	**Laptop-PT**	Admin PC	202	172.30.MN.19	/28 (.240)	172.30.MN.17	172.30.MN.17	Yellow
Company 1	**Company 1-Firewall**	C1-Edge	201	172.30.MN.3	/29 (.248)	172.30.MN.2	-	Black
Company 1	**Company 1-Firewall**	C1-LAN GW	110	172.30.MN.49	/28 (.240)	-	-	Brown
Company 1	**C1-LAN-Server**	App Server	110	172.30.MN.50	/28 (.240)	172.30.MN.49	172.30.MN.49	Brown
Company 1	**Company 1-Firewall**	C1-DMZ GW	130	172.30.MN.65	/28 (.240)	-	-	Purple
Company 1	**Web Server**	WWW-01	130	172.30.MN.66	/28 (.240)	172.30.MN.65	8.8.8.8	Purple
Company 1	**C1-SAN-Switch**	iSCSI SW	140	172.30.MN.81	/28 (.240)	-	-	Green
Company 1	**C1-SAN-Server**	Storage	140	172.30.MN.82	/28 (.240)	172.30.MN.81	-	Green
Company 2	**Company 2-Firewall**	C2-Edge	201	172.30.MN.4	/29 (.248)	172.30.MN.2	-	Black
Company 2	**Company 2-Firewall**	C2-LAN GW	210	172.30.MN.97	/28 (.240)	-	-	Red
Company 2	**C2-LAN-Server**	App Server	210	172.30.MN.98	/28 (.240)	172.30.MN.97	172.30.MN.97	Red
Company 2	**Company 2-Firewall**	C2-DMZ GW	230	172.30.MN.113	/28 (.240)	-	-	Pink
Company 2	**Web Server**	WWW-02	230	172.30.MN.114	/28 (.240)	172.30.MN.113	8.8.8.8	Pink
Company 2	**C2-SAN-Switch**	iSCSI SW	240	172.30.MN.129	/28 (.240)	-	-	Orange
Company 2	**C2-SAN-Server**	Storage	240	172.30.MN.130	/28 (.240)	172.30.MN.129	-	Orange

9. Added values-TBD

Site 1 (Private Cloud):

- Security Logon Banner: We will configure a Group Policy Object (GPO) to display an "Interactive Logon Message" warning (e.g., Authorized Use Only) before login. This ensures compliance with enterprise security standards by legally advising users of acceptable usage policies.
- Disk Quota Management: To prevent storage exhaustion, we will enforce strict Disk Quotas on the file servers. This limits the amount of storage space available to each user, ensuring fair resource distribution and preventing abuse.
- Standardized Corporate Desktop: We will deploy a customized corporate desktop background via GPO to all client workstations. This enforces a consistent professional identity and demonstrates the ability to manage user environments centrally.

Site 2 (Public Cloud):

- Customized Web Branding: We will modify the default web server configuration to display a custom "Company Portal" landing page with team branding. This simulates a real-world corporate intranet environment rather than a default installation.
- Client System Hardening: We will implement strict Group Policies to restrict user access to critical system configurations, such as disabling the Control Panel and right-click context menus. This "kiosk-style" locking prevents unauthorized changes to the client OS.
- ZFS Data Redundancy: Utilizing the TrueNAS deployment at Site 2, we will configure ZFS Mirroring for the storage pools. This provides enterprise-grade data integrity and protects cloud-hosted data against corruption or virtual disk failure.

10. List of Attachments

Note: the following Excel files are attached to this submission as integral parts of the project plan.

Please refer to the specific files below for full details:

- **Attachment A: Project Schedule & Test Plan**
  - **File Name:** Gantt Chart.xlsx
  - **Description:** This file contains the master **Gantt Chart** timeline, team role assignments, and the comprehensive **Test Plan** (including test phases, strategies, and schedules).
- **Attachment B: Network Configuration**
  - **File Name:** NetworkTopology.xlsx
  - **Description:** This file includes the **Network Topology Diagram** and the detailed **IP Address Allocation Tables** for both Site 1 (Private Cloud) and Site 2 (Public Cloud).
- **Attachment C: Hardware Specifications**
  - **File Name:** Hardware Plan.xlsx
  - **Description:** This file details the hardware resource planning, including vCPU, RAM, and Storage specifications for all virtual machines and infrastructure components.